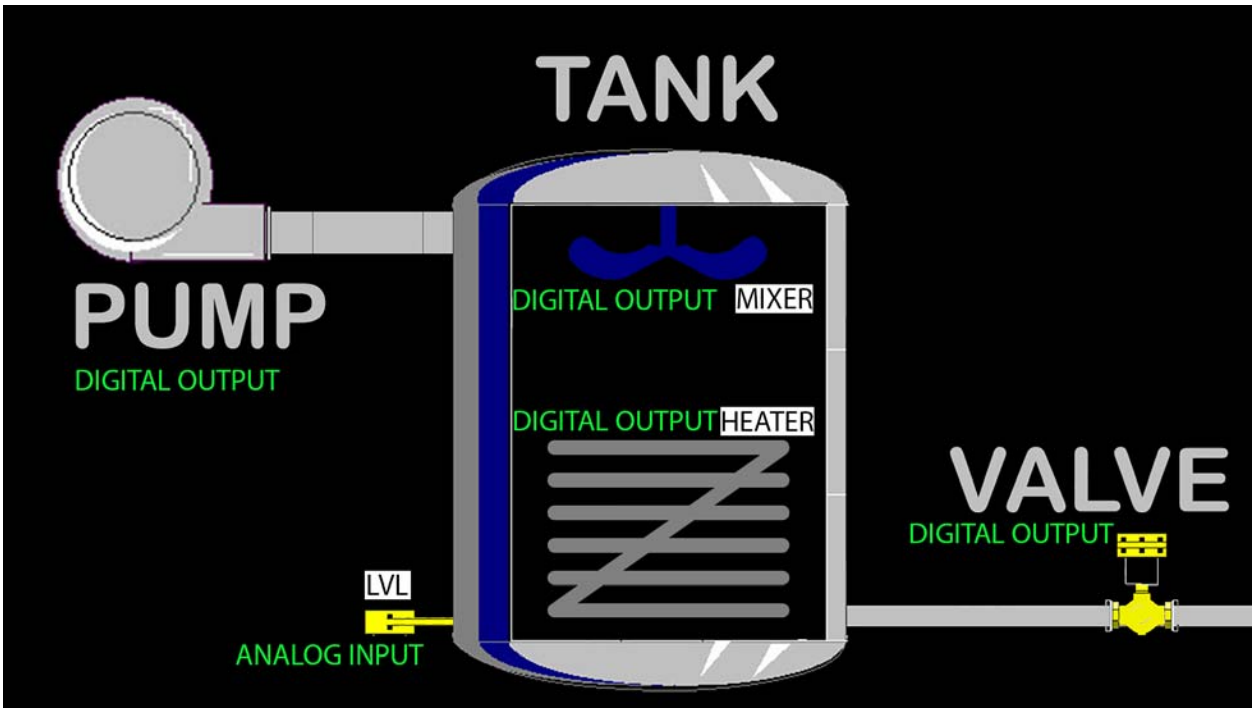
Project 7



Machine has the following stages (SFC):

0 - OFFLINE (LD) - All devices are shutdown.

1 - INIT (FBD) - Once activated, it initializes. This resets all values and prepares the for operation.

2 - STARTUP (ST) - After initialization, the machine warms up. First the tank is filled to between 40% and 70% and then the heater brings the temperature up to between 300 and 350 degrees. Once temperature is attained, the mixer (MIXER) starts running are remains running until the temperature falls below 300 degrees. Mixer must run for 30 seconds before machine can operate (next stage)

3 - OPERATION (CFC) - Valve opens and depletes the tank constantly. Pump runs as needed filling tank and lowering temperature. Tank must be maintained between 20% and 80%. Temperature must be maintained between 400 and 500 degrees via the heater.

4 - STABILITY (IL) - An operating fault (LL / HH level or HH temp) puts us into a corrective "stability" mode where the machine tries to fix itself before automatically returning to normal operation.

NOTES:

1) MODE (dint) must be updated 0-4 upon entering each stage

2) Alarms (LL_LVL, HH_LVL, HH_TMP) are self-resetting and can only occur in modes 2-3 (but will still exist in mode 4). Alarms will both trigger and reset on a 10-sec delay.

3) Level and Temperature will both come in as PID signals (0-16383) and be scaled as follows: LVL 0-100, TEMP 0-1000. A simulation routine written in the paradigm of your choosing will simulate the changes of these values over time. Temperature should semi-slowly try to fall to 70 (but never lower). With the pump running, the temperature should fall more quickly (but still manageably). Level should fall when valve is open (TRUE = open). Pump should fill faster than valve diminishes. Likewise, heater (HEATER) heats faster than pump causes temperature to fall. Level and temp changes will be applied every 500ms.

4) Pump, heater, mixer and valve will all energize and deenergize on a 5-sec delay.

5) We will count how many times we enter mode 2. We will count all of the time we spend in mode 4 but we will only update our HRS, MINS and SECS (DINTs) right before transitioning out.

6) Four DINTs (HEATUP, DROPHEAT, FILLRATE and DRAINRATE) will be used to govern the rate of change for level and temperature. We will be adding / subtracting these against out analog inputs in the simulation routine.

7) There are NO test criteria and no, I won't be grading / reviewing these. If you can make it work according to the above description and to your own satisfaction, congratulations: you're a programmer!

General functionality: Anything that is programmed independently of the stages can be programmed in the paradigm of your choosing.

Cheats: If you HAVE to cheat on anything that you can't figure out how to do within the designated paradigm, create an IL program, and do it in there.